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INSURANCE POLICY

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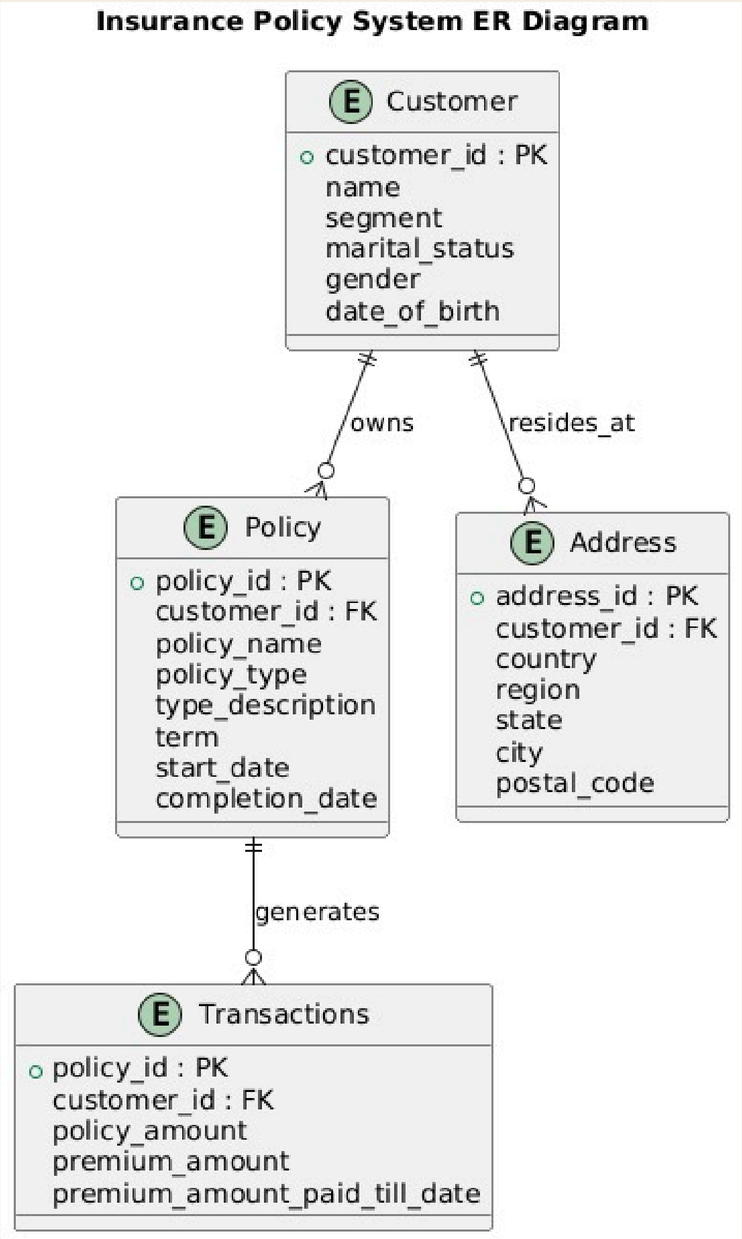
PROBLEM STATAEMENT

- ABC Insurance collects customer, policy, and transaction data from multiple regions in the US.
- The data must be ingested, cleaned, and stored in a centralized data warehouse.
- A data pipeline will standardize and transform the data.
- The system will calculate late fees and policy installments.
- It will also support analytics like policy changes and region-wise policy amounts.

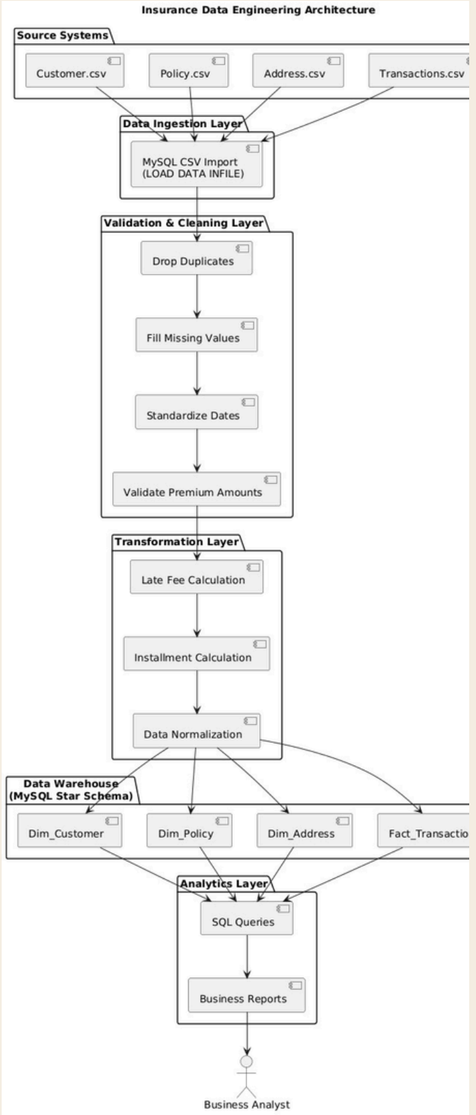
E-R DIAGRAM

- The ER diagram shows four main tables: Customer, Policy, Address, and Transactions.
- A customer can have multiple policies.
- Each customer is connected to an address showing their region and location.
- Every policy has transactions that store premium and payment details.
- These relationships help store and analyze insurance data easily.



SYSTEM DESIGN

- Insurance data comes from CSV files like Customer, Policy, Address, and Transactions.
- These files are imported into MySQL to store the data.
- The data is then cleaned and validated by fixing errors, removing duplicates, and standardizing formats.
- After that, business calculations like late fees and installments are applied.
- Finally, SQL queries and dashboards are used to analyze the data and generate insights.



WORK OVERALL

- We created the raw data and cleaned it using Excel.
- The cleaned data was imported into MySQL.
- We used SQL to run the required queries.
- Finally, we created a dashboard in Tableau to show insights.

WORK

- Created raw CSV datasets for customer/candidate, policy, transactions, and address data.
- Cleaned the data by removing duplicates, handling missing values, and organizing it for analysis.
- Designed an ER diagram and system architecture to represent data relationships and flow.
- Imported the datasets into MySQL and created tables with proper keys and relationships.
- Reduced the dataset from 1 lakh records to 100 records to enable faster query testing.
- Wrote 6–7 SQL queries to solve the required business problems and extract insights.
- Built Tableau dashboards to visualize and present the final insights clearly.

PROJECT TIMELINE

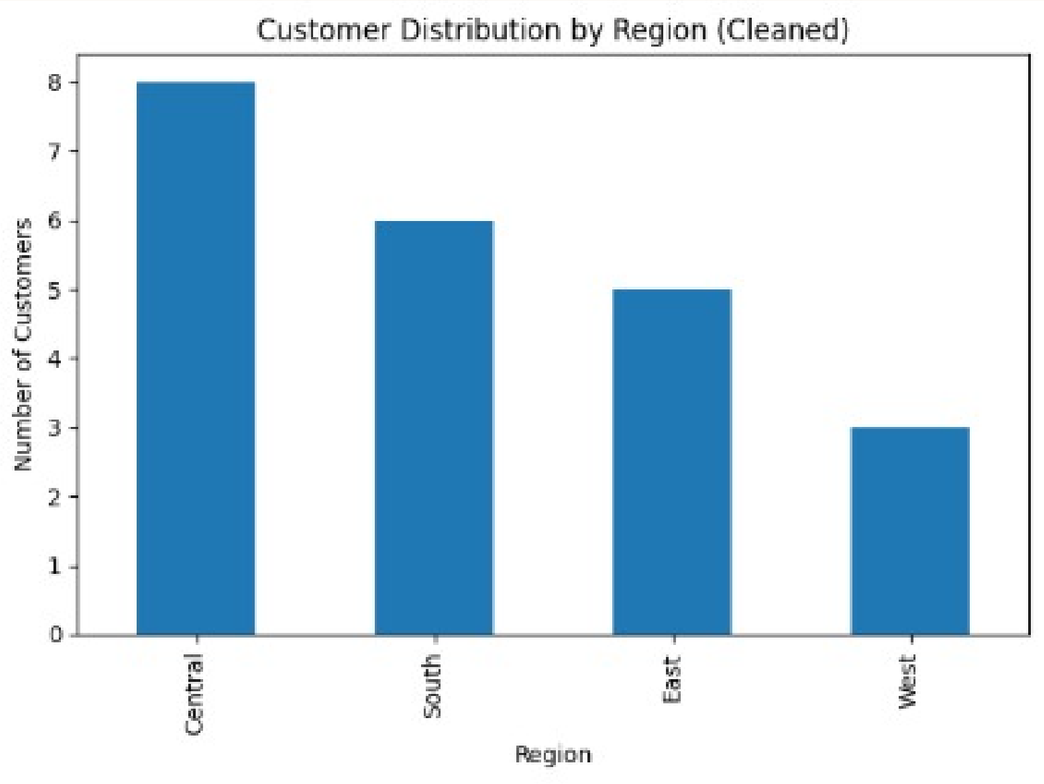
DESIGN - 1Hr

IMPLEMENTATION/TESTING - 3-4 Hrs

DEMO - 1-2 Hrs

CUSTOMER DISTRIBUTION BY REGION

This chart shows how customers are distributed across different geographical regions.
It helps identify which regions have the highest number of customers and where the insurance company has stronger market presence.
Regions with fewer customers may indicate potential areas for business expansion.

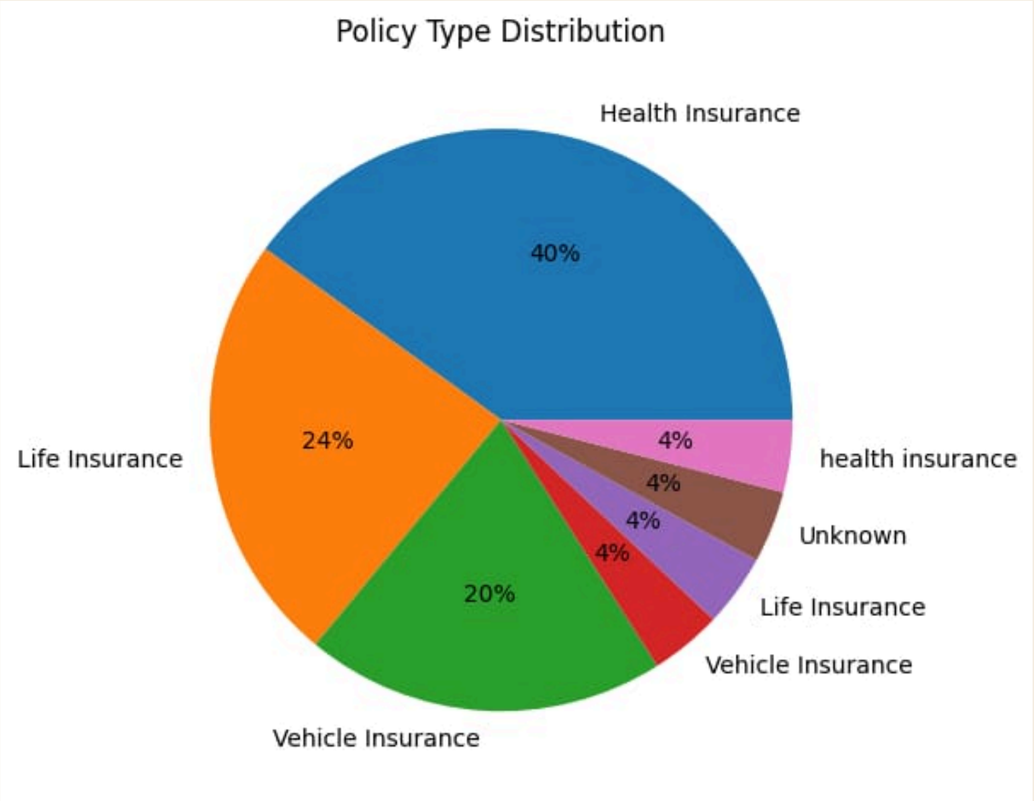


POLICY TYPE DISTRIBUTION

This visualization shows the proportion of different policy types such as Health, Life, and Vehicle insurance.

It helps understand which insurance products are most popular among customers.

The company can use this insight to focus marketing efforts on high-demand policy types.

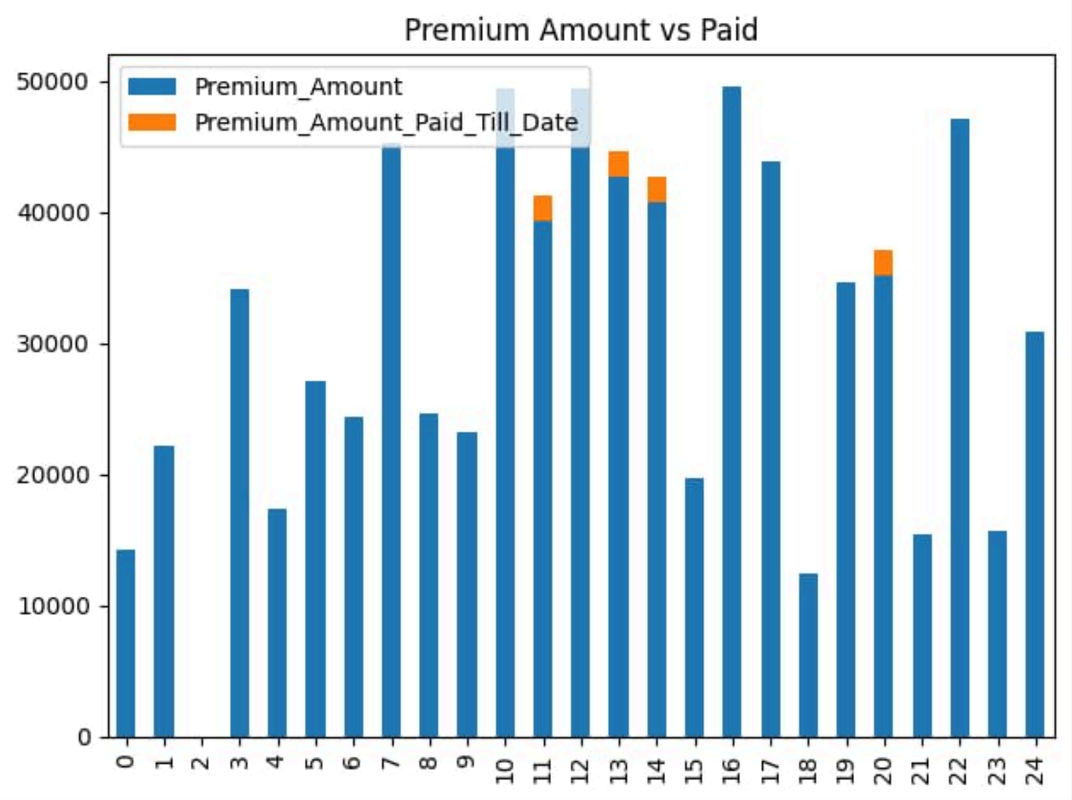


PREMIUM AMOUNT VS PREMIUM PAID TILL DATE

This chart compares the total premium amount with the amount already paid by customers.

It helps monitor payment progress and identify potential delays in premium payments.

This insight can help insurance companies track revenue flow and detect overdue payments.



WORK DISTRIBUTION

- Use case selection: The team first selected an appropriate problem statement and use case for the project.
- Raw dataset extraction: Dhilip collected and prepared the raw dataset from the source files.
- Raw to clean data processing: Shreya cleaned the raw data by removing duplicates, fixing missing values, and standardizing formats before importing it into the database.
- ER diagram & system design: Krishna and Shreya designed the ER diagram and overall system architecture.
- Presentation document: Shreya prepared the presentation content and documentation.
- Data import to MySQL: Dhilip imported the cleaned data into MySQL to run SQL queries with the help of the team.
- Visuals & final presentation: Shreya and Krishna created visualizations using Tableau and finalized the presentation.

CONCLUSION

- Combined customer, policy, address, and transaction data from different regions into one centralized data warehouse.
- Built a data pipeline to load the CSV files, clean the data, and standardize the formats.
- Performed data transformations such as calculating late fees and policy installments.
- Organized the data using fact and dimension tables to make querying easier.
- Used SQL queries and Tableau dashboards to generate insights like policy changes and region-wise policy amounts.

THANK YOU

Thanks For Watching